ICQS

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"Think Different, Build Different: Just Do It"

VIKI

Variational Intelligence for Key Insights

Satellite anomalies don't wait. Neither should your operators.

Your satellite generates thousands of telemetry readings every minute. Hidden inside that data are early warning signals — subtle deviations that, left undetected, become mission-critical failures. VIKI finds them automatically, so your operators can act before problems escalate.

WHY VIKI?

Built for operators, not data scientists — No specialist knowledge required. VIKI presents anomalies in plain language — which channels are affected, how severe, and when it started.

Works with any satellite, any ground segment — Upload your telemetry, answer four simple questions, and VIKI configures itself. No integration project. No proprietary hardware. On-premise, cloud, or air-gapped.

Scales with your constellation — One satellite or a full constellation — VIKI scales from lightweight single-server to distributed infrastructure.

Detects correlated faults across channels — When multiple subsystems fail together, VIKI groups the events into a single coherent alert. Less noise, faster operator response.

Space Domain Awareness ready — Channel attribution maps directly to RSO monitoring — identifying unexpected manoeuvres, emission changes and signature shifts.

Future-ready architecture — Designed to interface with quantum computing platforms. Civilian and defence capabilities cleanly separated.

THREE INTERFACES — ONE WORKFLOW

Training UI — Upload telemetry, answer 4 questions, VIKI trains and deploys. Expert parameter control available.

Level 2 — Monitoring — Real-time dashboards: system health, anomaly timelines, channel reconstruction error.

Level 3 — Event Management — Tickets, notes, event correlation and closure. Full audit trail for shift handover.

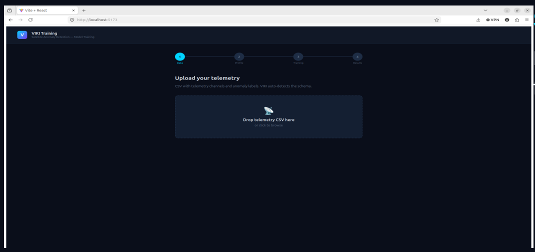
INDEPENDENTLY VALIDATED AGAINST ESA/ESOC STANDARDS

Tested against the ESA/ESOC Anomaly Detection Benchmark — real satellite telemetry used by the European space operations community.

F1C	PRECISION	RECALL	DATASET
99.8%	**99.6%**	**100%**	3-month / Clustered
88.6%	**93.7%**	**84%**	5-month / Sparse
ESA target ≥ 88%	ESA target ≥ 90%	ESA target ≥ 85%	

VIKI Training UI

From raw telemetry to deployed anomaly detection — four steps, no specialist knowledge required.

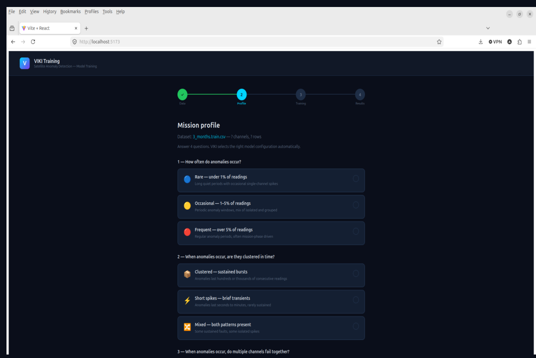


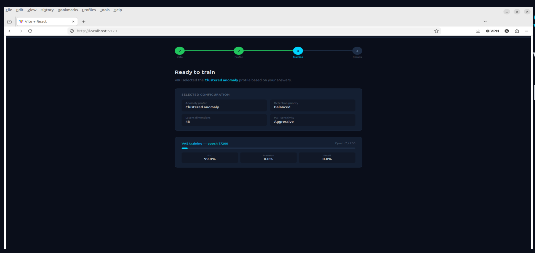
Step 1 — Upload your telemetry

Drag and drop your satellite telemetry file. VIKI automatically identifies sensor readings, fault indicators and timestamps — no manual mapping required.

Step 2 — Describe your mission

Answer four plain-language questions about anomaly behaviour. VIKI selects the right detection strategy automatically — no data science degree required.



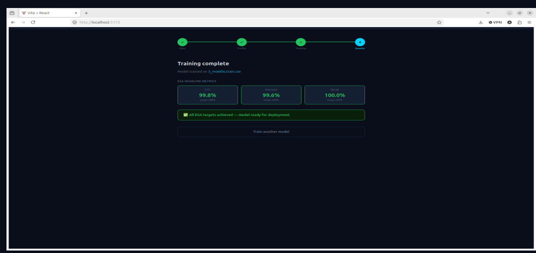


Step 3 — VIKI trains automatically

VIKI learns your satellite's normal behaviour and calibrates detection thresholds. Live progress indicator. No operator input needed.

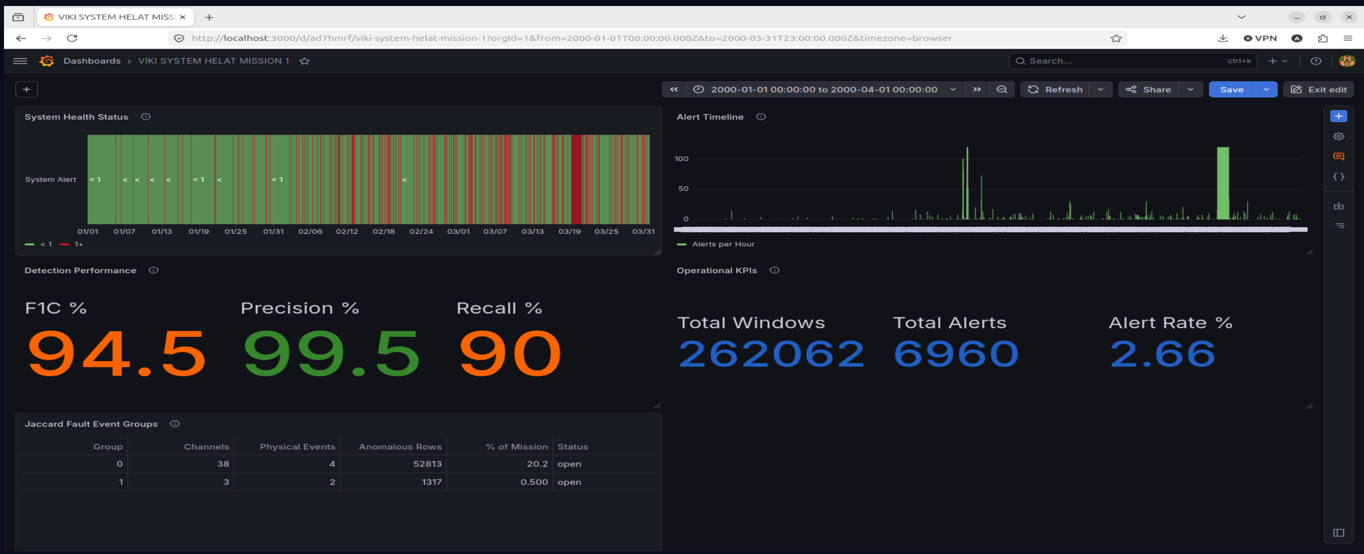
Step 4 — Results & one-click deployment

F1C 99.8%, Precision 99.6%, Recall 100% — all ESA targets achieved. One click deploys the model live to monitoring and event management.



VIKI Level 2 — Mission Monitoring

Real-time situational awareness — from system health to individual channel behaviour.

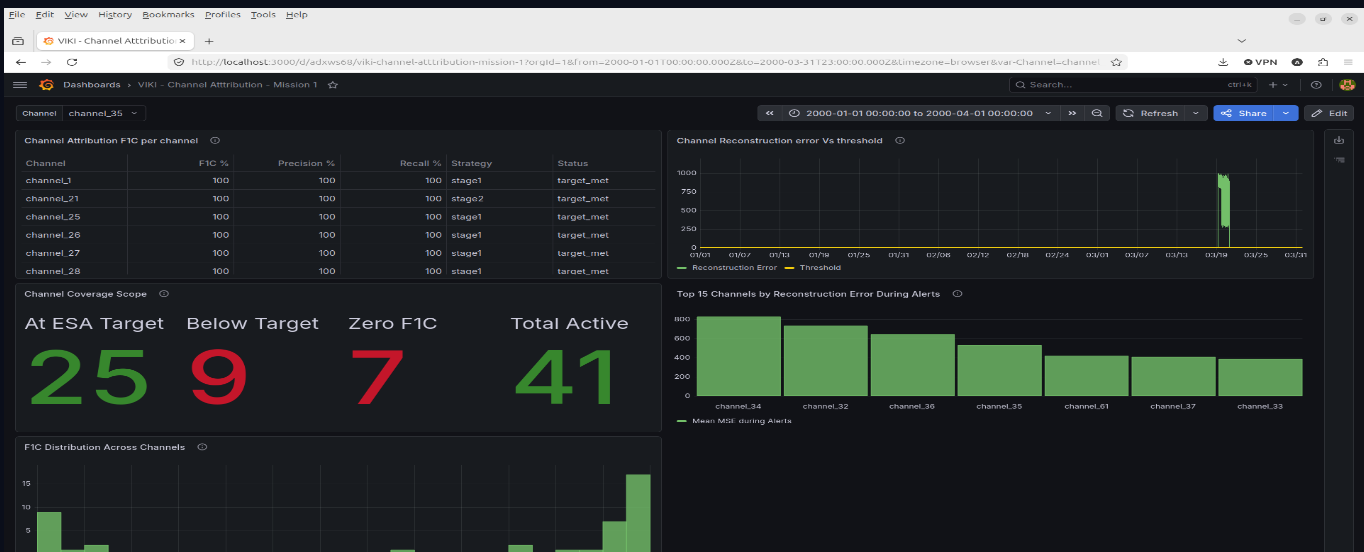


■ **System Health Status** — Colour-coded mission timeline — green for normal, red for anomaly events. Operators see when problems occurred and how long they lasted, without reading thousands of log entries.

■ **Alert Timeline** — Alerts-per-hour chart across the mission. Spot quiet vs high-activity windows instantly and correlate anomalies with mission events — manoeuvres, eclipse periods, payload activations.

■ **Detection Performance & KPIs** — Live F1C, Precision and Recall plus total windows, alerts and alert rate. Mission management sees VIKI's real-time detection quality with no technical background needed.

■ **Jaccard Fault Event Groups** — When multiple channels fail together, VIKI groups them into correlated fault events — showing which subsystem groups are active and what percentage of mission time they cover.



■ **Channel Attribution F1C** — Detection accuracy ranked per channel — F1C, Precision, Recall and strategy. Channels marked 'target_met' exceed ESA thresholds. Operators see which subsystems are most reliably covered.

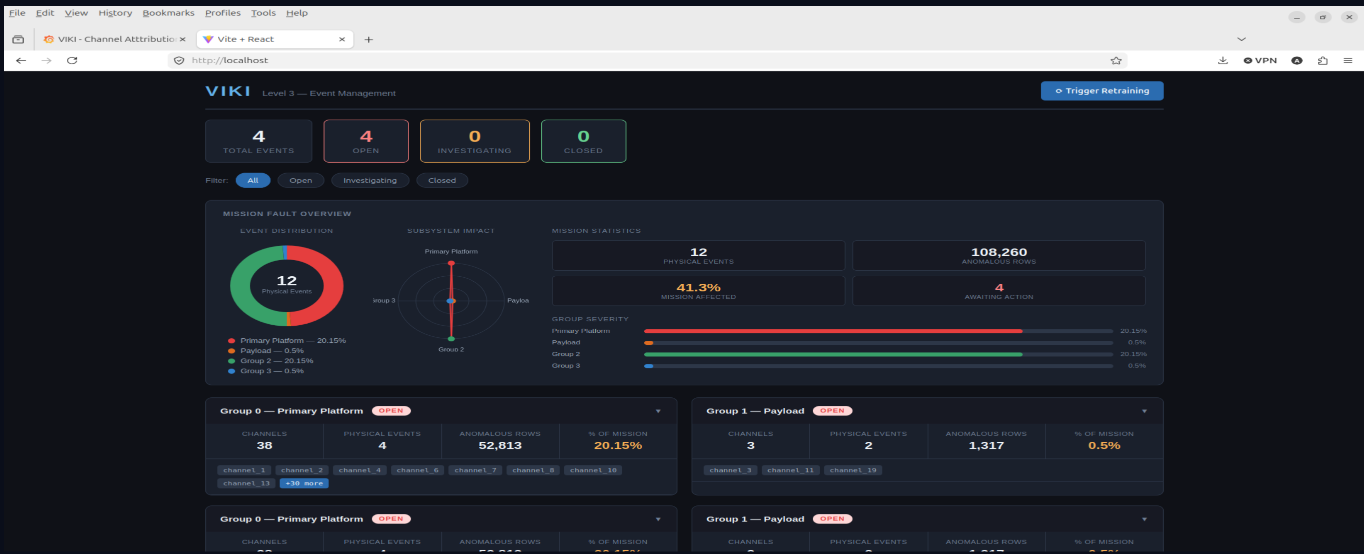
■ **Reconstruction Error vs Threshold** — The core detection signal — channel deviation from normal vs the adaptive threshold. When the signal spikes above the line, VIKI raises an alert. The 19 March anomaly is clearly visible.

■ **Top 15 Channels During Alerts** — During anomaly events, ranks channels by deviation — telling operators which subsystems to investigate first. Reduces investigation time from hours to minutes.

■ **Channel Coverage Scope** — At-a-glance: channels at ESA target, below target, or with insufficient data. Gives mission managers an instant health-check of VIKI's full telemetry coverage.

VIKI Level 3 — Event Management

Where anomaly detection becomes operational action — built for Mission Control shift teams.

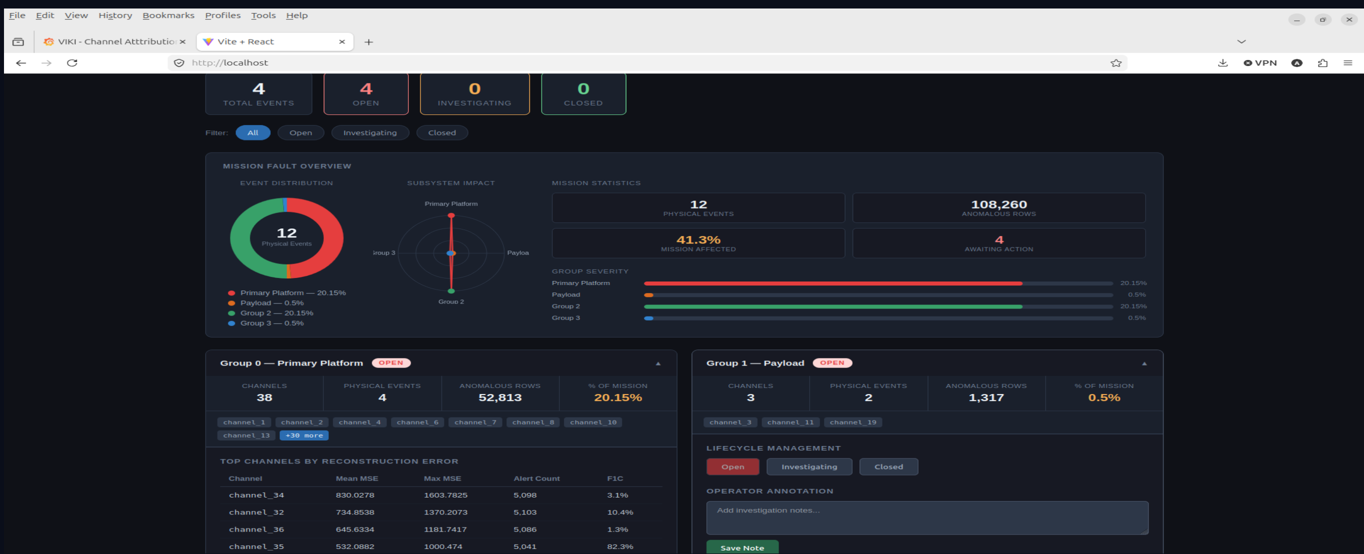


■ **Event Status Dashboard** — At the top of every shift, operators see exactly how many events are open, under investigation, and closed — no need to search logs or query databases. The mission state is visible at a glance.

■ **Mission Fault Overview** — Visual breakdown by subsystem — Event Distribution pie, Subsystem Impact radar and Group Severity bars. Operators immediately understand which parts of the satellite are most affected.

■ **Correlated Fault Event Groups** — VIKI clusters channels that fail together into named groups. Each group shows channels involved, physical events, anomalous rows and mission time affected — turning hundreds of alerts into a handful of actionable events.

■ **Mission Statistics** — Physical events detected, anomalous rows, mission-affected percentage and events awaiting action — giving mission management a complete operational picture in four numbers.



■ **Top Channels by Reconstruction Error** — Inside each fault group, VIKI ranks channels by severity — Mean MSE, Max MSE, alert count and F1C. Operators go straight to the most affected channel without manually reviewing each one.

■ **Lifecycle Management** — Each fault event moves through Open → Investigating → Closed with a single click. The entire team sees current state in real time — critical for multi-shift operations and handover briefings.

■ **Operator Annotation** — Operators add investigation notes directly to each fault event — recording findings, actions taken and what to watch next shift. Timestamped and stored permanently for mission reporting and lessons-learned.

■ **Trigger Retraining** — When new anomaly patterns emerge, operators trigger model retraining directly from Level 3 — without leaving the operational environment or involving a data science team.